Lab 2 Activity 2.1 - Configuring a small Network

Online: Not assessed

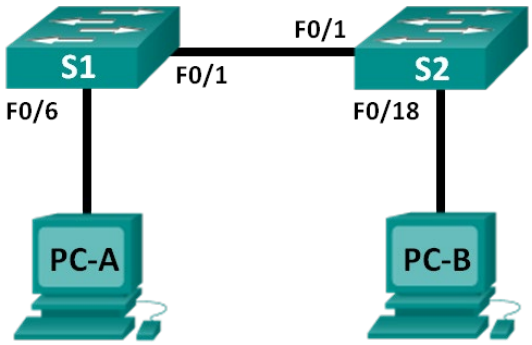
You must use Packet Tracer v8.2.2 to complete this lab.

Please rename your Packet Tracer file to FirstName_LastName_sNumber_Lab2.1.pka

Scoring Scheme

No.	Description	Points
1	PC configuration	10
2	Switch configuration	38

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
PC-A	NIC	192.168.1.10	255.255.255.0
PC-B	NIC	192.168.1.11	255.255.255.0
S1	VLAN 1	192.168.1.12	255.255.255.0
S2	VLAN 2	192.168.1.13	255.255.255.0

Objectives

Part 1: Set Up the Network Topology (Ethernet only)

Part 2: Configure PC Hosts

Part 3: Configure and Verify Switch Settings

Part 4: Submit Packet Tracer file

Background / Scenario

In this lab, you will build a simple network using Ethernet LAN cabling and access Cisco switches using the console and remote access methods. You will configure basic switch settings, IP addressing, and demonstrate the use of a management IP address for remote switch management. You will also configure basic settings including hostname, local passwords, and login banner. Use **show** commands to display the running configuration, IOS version, and interface status.

You will apply IP addressing for this lab to the PCs to enable communication between these two devices. You do not need to calculate the IP addresses as these will be provided. You will use the **ping** utility to verify connectivity.

For remote management, Cisco switches have a special interface, known as a Switch Virtual Interface (SVI). The SVI can be configured with an IP address, commonly referred to as the management address. The management address is used for remote access to the switch to display or configure settings.

Note: Make sure that the switches have been erased and have no startup configurations. Refer to Appendix A for the procedure to initialize and reload a switch.

Required Resources

- Packet Tracer 8.2.2
- 2 Packet Tracer Cisco 2960 Switches
- 2 Packet Tracer PCs
- Ethernet cables as shown in the topology

Part 1: Set Up the Network Topology

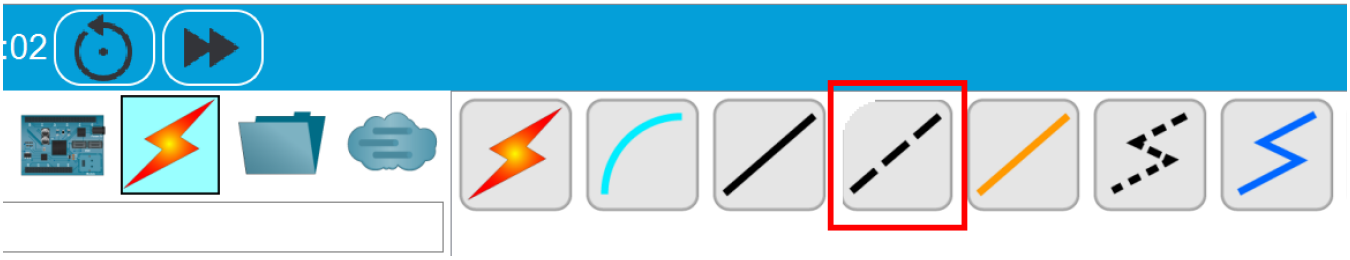
In this part you will complete the Packet Tracer activity. This Packet Tracer activity is not assessed. You must use the supplied Packet Tracer file for Lab 2 to complete the workshop as otherwise you will not be able to use the built in feed-back and scoring system that will also be used for later assessments. Using the supplied Packet Tracer file, you will cable the devices together according to the network topology.

Step 1: Create the devices.

Create all devices in the topology using Packet Tracer. Use the 2960 Switch and PC as end-devices. Update the Packet Tracer display names for the devices based on the topology.

Step 2: Connect the two switches.

Connect one end of a Cross-over Ethernet cable to F0/1 on S1 and the other end of the cable to F0/1 on S2. The cross-over cable is made of dashed lines in Packet Tracer and is used to connect two similar devices like switches together. It is identified in Packet Tracer in the image below:



You can connect a cable from one device to the next by using the cable menu (shown above), then:

- left click on the cable type (Crossover cable in the example above)
- left click on device 1
- choosing the port to connect to from the pop-up menu
- move the mouse (and the cable) to the second device
- left-click on the second device

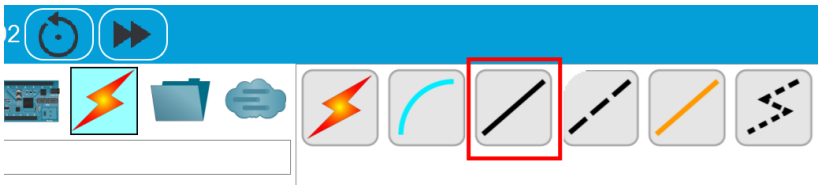
- finally choose the port that you want to connect to on the second device.

You should see the lights for F0/1 on both switches turn amber and then green. This indicates that the switches have been connected correctly.

Step 3: Connect the PCs to their respective switches.

a. Connect one end of an Ethernet cable to the NIC port on PC-A. In Packet Tracer this will be port FastEthernet 0. The type of cable that is required is the Copper Straight-Through cable.

   The Copper Straight-Through cable is made of a single solid line in Packet Tracer and is used to connect an end device to an intermediate device like a switch. It is identified in Packet Tracer in the image below:



   Connect the other end of the cable to F0/6 on S1. After connecting the PC to the switch, you should see the light for F0/6 turn amber and then green, indicating that PC-A has been connected correctly.

b. Connect one end of an Ethernet cable to the NIC port on PC-B. In Packet Tracer this will be port FastEthernet 0. Connect the other end of the cable to F0/18 on S2. After connecting the PC to the switch, you should see the light for F0/18 turn amber and then green, indicating that the PC-B has been connected correctly.

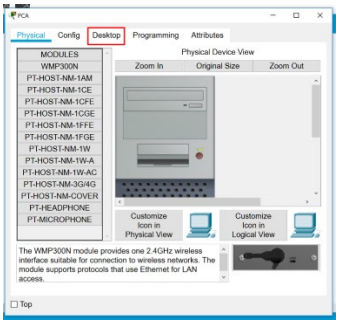
Step 4: Visually inspect network connections.

After cabling the network devices, take a moment to carefully verify the connections to minimize the time required to troubleshoot network connectivity issues later.

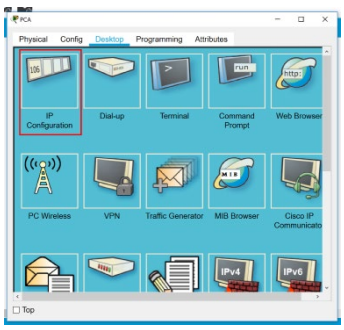
Part 2: Configure PC Hosts

Step 1: Configure PC settings.

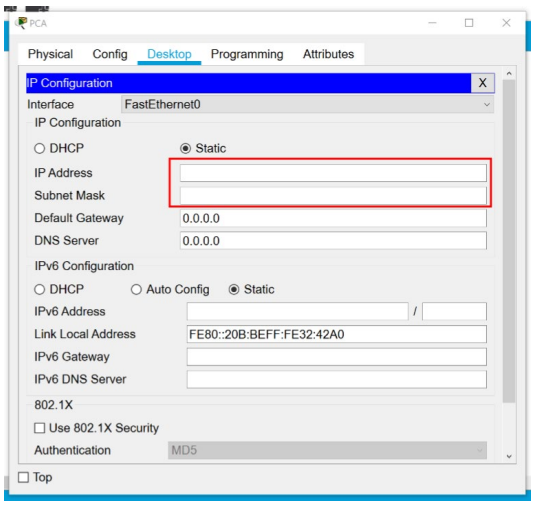
Left click on PC-A and select the Desktop tab:



Then select IP Configuration:



Enter the IP address and subnet mask as specified in the Topology:

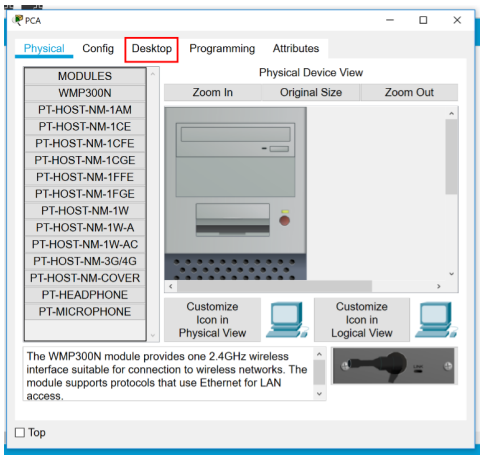


Repeat the previous steps to enter the IP address information for PC-B.

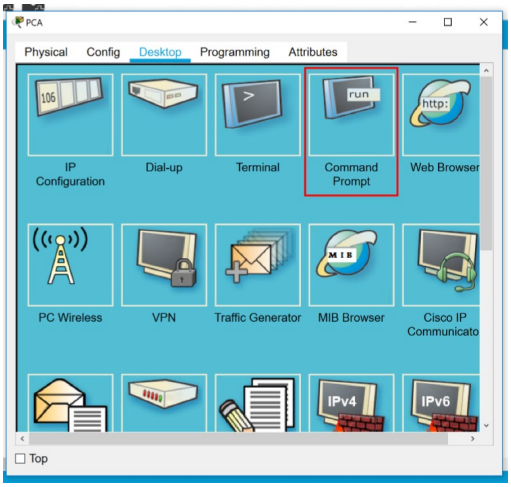
Step 2: Verify PC settings and connectivity.

Before continuing, check that there is connectivity between PC-A and PC-B.

Left click on PC-A and select the Desktop tab:

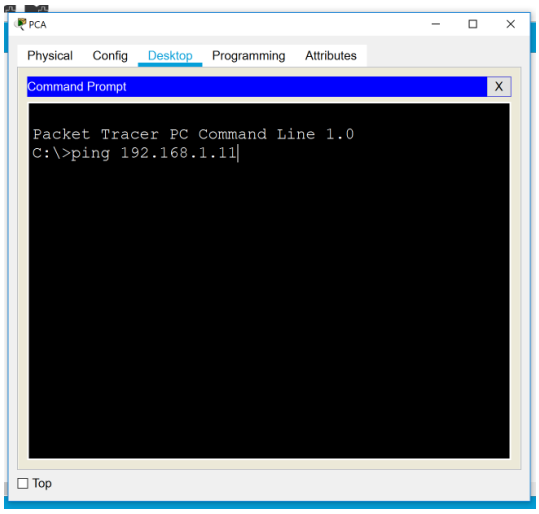


Then select Command prompt:

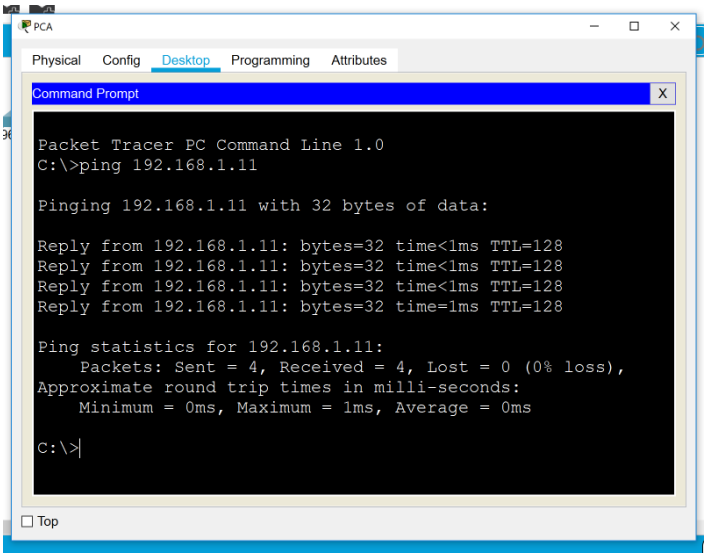


Now use the ping command to check the connectivity between PC-A and PC-B. The IP address for PC-B is 192.168.1.11

In the command prompt window, type **ping 192.168.1.11** and press Enter:



You should see a result something like this (note that the first ping may be unsuccessful if the network is still resolving the connectivity between devices):



The information in the above ping result shows that there were 4 ping connectivity requests sent to PC-B (192.168.1.11) and all 4 of them resulted in a response from PC-B (0% loss).

In your Packet Tracer network, were the ping results successful? _____

If not, troubleshoot as necessary.

Note: If you did not get a reply from PC-B, try to ping PC-B again – sometimes it takes a little while for the network connectivity to resolve. If you still do not get a reply from PC-B, try to ping PC-A from PC-B. If you are unable to get a reply from the remote PC, then ask your workshop instructor during your lab or ask for assistance during the common time.

Part 3: Configure and Verify Switch Settings - Console

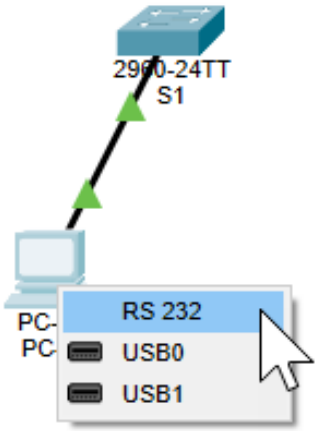
Step 1: Connect a console cable from PC-A to Switch S1

To connect to the switch from PC-A, you will need to connect a console cable from the RS232 port on the PC to the console port on the switch.

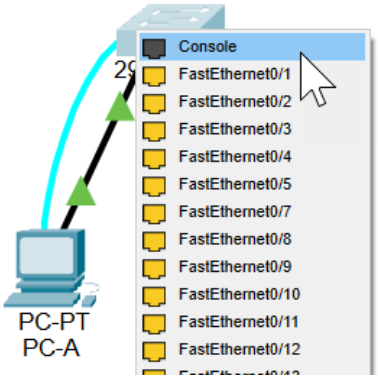
The type of cable that is required is the Console cable.

The Console cable is made of a single solid blue line in Packet Tracer and is used to connect an end device to an intermediate device like a switch so that commands can be sent between them. It is identified in Packet Tracer in the image below:

a. Connect one end of a Console cable to the RS232 port on PC-A. Do this by left-clicking on the blue cable icon at the bottom of the Packet Tracer screen, then left clicking on PC-A to bring up the available ports. In Packet Tracer we will be connecting the console cable to the RS322 (serial connection) port.



b. Connect the other end of the Console cable to the console port on the Switch.



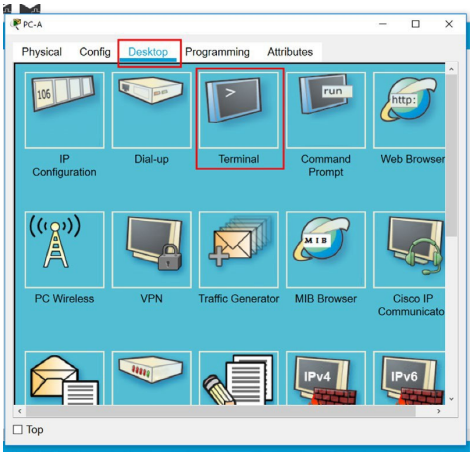
Step 2: Use the terminal program to make a console connection between PC-A and Switch S1

In this step, you will configure basic switch settings.

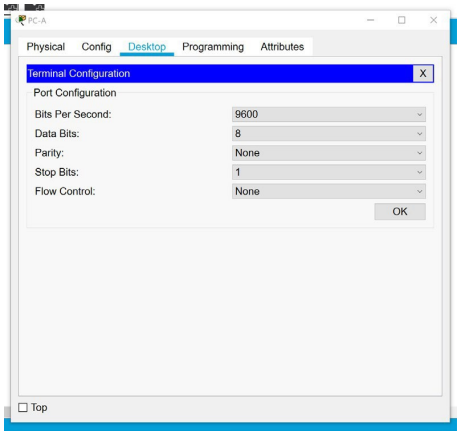
a. Use the terminal program on PC-A in Packet Tracer to connect to the switch S1. This connection will use the console cable that connects PC-A and S1.

b. On the Terminal configuration window press OK. This will open the Switch Command Line Interface (CLI).

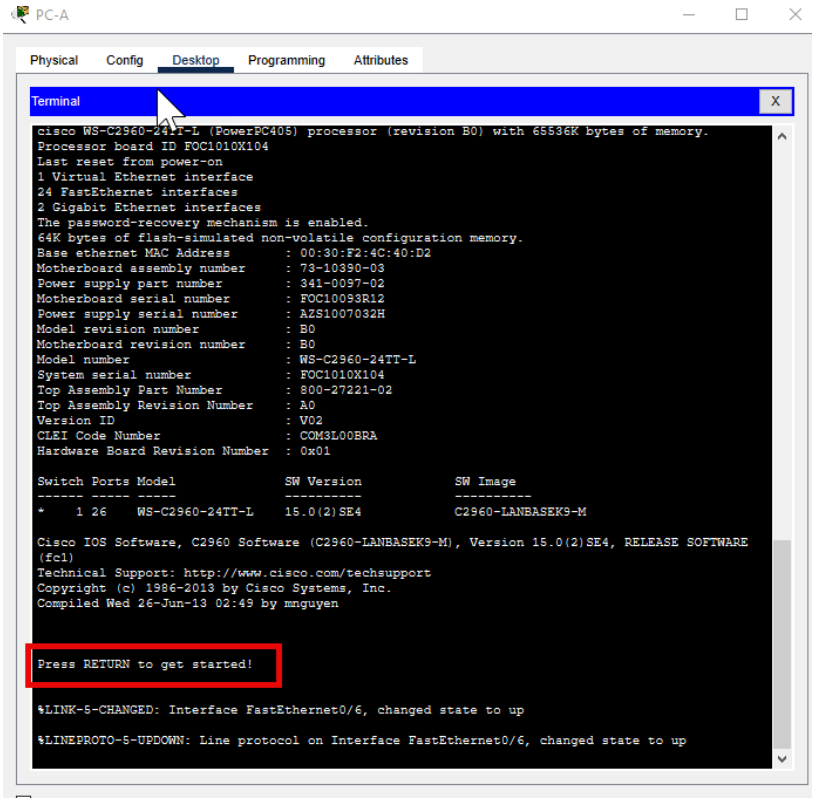
c. You will see a terminal window where you will be able to enter commands from PC-A directly to the switch S1. Note that you will need to press the 'Return' or 'Enter' key to bring up the prompt before you can enter commands.



(a)



(b)



(c)

Step 3: Enter privileged EXEC mode.

You can access all switch commands in privileged EXEC mode. The privileged EXEC command set includes those commands contained in user EXEC mode, as well as the **configure** command through which access to the remaining command modes are gained. Enter privileged EXEC mode by entering the **enable** command.

```
Switch> enable
Switch#
```

The prompt changed from **Switch>** to **Switch#** which indicates privileged EXEC mode.

Step 4: Enter configuration mode.

Use the **configuration terminal** command to enter configuration mode.

```
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

Note: The prompt changed to reflect global configuration mode. You can use '**exit**' to go back one level, or '**end**' to go back to privileged exit mode directly. In this case **exit** would take you back to the privileged exec mode from Step 2 with prompt: `Switch#`

Step 5: Give the switch a name.

Use the **hostname** command to change the switch name to **S1**.

```
Switch(config)# hostname S1
S1(config)#
```

Step 6: Prevent unwanted DNS lookups.

To prevent the switch from attempting to translate incorrectly entered commands as though they were hostnames, disable the Domain Name System (DNS) lookup.

```
S1(config)# no ip domain-lookup
S1(config)#
```

Step 7: Enter local passwords.

To prevent unauthorized access to the switch, passwords must be configured and a login is required on the console line (con 0).

Set the enable secret password to restrict access to the commands when already logged into the system:

```
S1(config)# enable secret class
```

Configure the console line password to restrict access using the console cable. To prevent the switch from interrupting you when typing commands, turn on synchronous logging messages from within the console line configuration (logging synchronous).

```
S1(config)# line con 0
S1(config-line)# password cisco
S1(config-line)# login
S1(config-line)# logging synchronous
S1(config-line)# exit
S1(config)#
```

The switch we are using has 16 Virtual Terminal lines (VTY) numbered from 0 to 15. Each of these lines can support a remote connection and each line must have a password attached. You can configure a range of VTY lines at once using the command: `line vty start end` (eg: `line vty 0 15` configures all 16 at once).

Configure the vty line passwords to restrict access during remote access. Use the same password that is used for console access.

```
S1(config)# line vty 0 15
S1(config-line)# password cisco
S1(config-line)# login
S1(config-line)# logging synchronous
S1(config-line)# exit
S1(config)#
```

Note: In the scoring for this item sometimes this will be worth 1 mark per switch and could be attached to only one of the 16 vty lines but please set the commands for all vty lines.

Step 8: Enter a login MOTD banner.

A login banner, known as the message of the day (MOTD) banner, should be configured to warn anyone accessing the switch that unauthorized access will not be tolerated.

The **banner motd** command requires the use of delimiters to identify the content of the banner message. The delimiting character can be any character as long as it does not occur in the message. For this reason, symbols, such as the **single quote '**, are often used.

Important: Please enter the following banner exactly as written. Using the correct and approved language is critical to ensure that legal requirements are met. If you change any character in the **Unauthorised access is prohibited and strictly prosecuted to the full extent of the law.** (including the final full-stop at the end) the banner will be marked as incorrect as it does not exactly meet the requirement:

```
S1(config)# banner motd 'Unauthorised access is prohibited and
strictly prosecuted to the full extent of the law.'
S1(config)# exit
S1#
```

Step 9: Configure the Switch Virtual Interface (SVI) for S1

Cisco switches have a special interface, known as a Switch Virtual Interface (SVI). The SVI can be configured with an IP address, commonly referred to as the management address. The management address is used for remote access to the switch to display or configure settings. So far, we have been connecting to the switch directly via the console cable when entering commands. This is not always possible if we are not physically close to the machine. Therefore, we also want to configure the switch to allow remote access and we will use the VLAN 1 interface with an IP address to achieve this.

In this step we will configure the SVI IP address using the VLAN 1 interface. Later in this lab we will connect to the switch remotely using a terminal program and connecting to the IP address attached to VLAN 1.

To configure the Switch Virtual Interface (SVI) on interface VLAN 1 with an IP address:

In global configuration mode set the SVI IP address to allow remote switch management.

```
S1# configure terminal
S1#(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.12 255.255.255.0
S1(config-if)# no shutdown
```

```
S1(config-if)# exit
S1(config)#
```

Note: The Switch Virtual Interface on Interface VLAN 1 will be shutdown (off) until we turn it on. The command to turn the interface on is `'no shutdown'`

Step 10: Display the current configuration

The **show running-config** command displays the entire running configuration, one page at a time. Use the spacebar to advance paging when you see the "- - More –" prompt. The commands configured in Steps 1 – 8 are highlighted below for S1. Also confirm for S2.

```
S1# show running-config
   Building configuration...

   Current configuration : 1380 bytes
   version 15.0
   no service timestamps log datetime msec
   no service timestamps debug datetime msec
   no service password-encryption
   !
   hostname S1
   !
   enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
   !
   no ip domain-lookup
   !
      <output omitted>
   !
   interface Vlan1
   ip address 192.168.1.12 255.255.255.0
   !
   banner motd ^CUnauthorised access is prohibited and strictly prosecuted
   to the full extent of the law.^C
   !
   line con 0
   password cisco
   logging synchronous
   login
   !
   line vty 0 4
   password cisco
   logging synchronous
   login
   line vty 5 15
   password cisco
   logging synchronous
   login
   !
   End
S1#
```

Step 11: S1#Display the IOS version and other useful switch information.

Use the **show version** command to display the IOS version that the switch is running, along with other useful information. Again, you will need to use the spacebar to advance through the displayed information.

```
S1# show version
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE, RELEASE SOFTWARE
(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Sat 28-Jul-12 00:29 by prod_rel_team

ROM: Bootstrap program is C2960 boot loader
BOOTLDR: C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(53r)SEY3, RELEASE SOFTWARE (fc1)

S1 uptime is 1 hour, 38 minutes
System returned to ROM by power-on
System image file is "flash:/c2960-lanbasek9-mz.150-2.SE.bin"


This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco WS-C2960-24TT-L (PowerPC405) processor (revision R0) with 65536K bytes of memory.
Processor board ID FCQ1628Y5LE
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.

64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address       : 0C:D9:96:E2:3D:00
Motherboard assembly number     : 73-12600-06
Power supply part number        : 341-0097-03
Motherboard serial number       : FCQ16270N5G
Power supply serial number      : DCA1616884D
Model revision number           : R0
Motherboard revision number     : A0
Model number                    : WS-C2960-24TT-L
System serial number            : FCQ1628Y5LE
Top Assembly Part Number        : 800-32797-02
```

```
Top Assembly Revision Number     : A0
Version ID                       : V11
CLEI Code Number                 : COM3L00BRF
Hardware Board Revision Number   : 0x0A


Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE             C2960-LANBASEK9-M

Configuration register is 0xF
S1#
```

Step 12: Display the status of the connected interfaces on the switch.

To check the status of the connected interfaces, use the **show ip interface brief** command. Press the spacebar to advance to the end of the list.

```
S1# show ip interface brief
    Interface              IP-Address      OK? Method Status                Protocol
    FastEthernet0/1        unassigned      YES manual up                    up
    FastEthernet0/2        unassigned      YES manual down                  down
    FastEthernet0/3        unassigned      YES manual down                  down
    FastEthernet0/4        unassigned      YES manual down                  down
    FastEthernet0/5        unassigned      YES manual down                  down
    FastEthernet0/6        unassigned      YES manual up                    up
    FastEthernet0/7        unassigned      YES manual down                  down
    FastEthernet0/8        unassigned      YES manual down                  down
    FastEthernet0/9        unassigned      YES manual down                  down
    FastEthernet0/10       unassigned      YES manual down                  down
    FastEthernet0/11       unassigned      YES manual down                  down
    FastEthernet0/12       unassigned      YES manual down                  down
    FastEthernet0/13       unassigned      YES manual down                  down
    FastEthernet0/14       unassigned      YES manual down                  down
    FastEthernet0/15       unassigned      YES manual down                  down
    FastEthernet0/16       unassigned      YES manual down                  down
    FastEthernet0/17       unassigned      YES manual down                  down
    FastEthernet0/18       unassigned      YES manual down                  down
    FastEthernet0/19       unassigned      YES manual down                  down
    FastEthernet0/20       unassigned      YES manual down                  down
    FastEthernet0/21       unassigned      YES manual down                  down
    FastEthernet0/22       unassigned      YES manual down                  down
    FastEthernet0/23       unassigned      YES manual down                  down
    FastEthernet0/24       unassigned      YES manual down                  down
    GigabitEthernet0/1     unassigned      YES manual down                  down
    GigabitEthernet0/2     unassigned      YES manual down                  down
    Vlan1                  192.168.1.12    YES manual up                    up
    S1#
```

Why are some FastEthernet ports on the switches up and others are down?

_____

_____

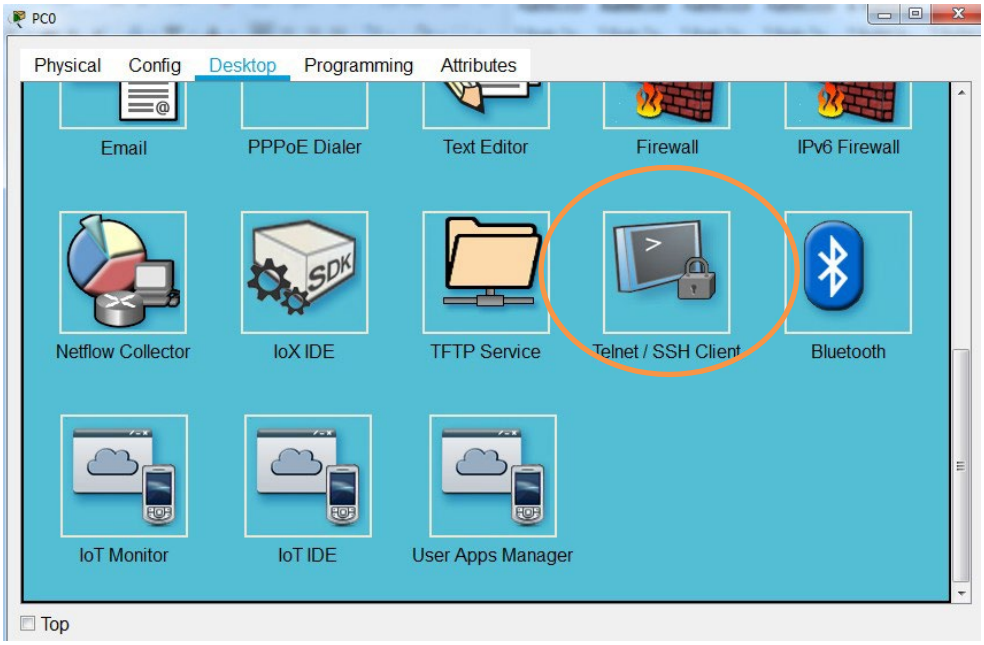
Step 13: Repeat Steps 1 to 12 to configure switch S2.

The only differences for this step are to connect PC-A to the console port of Switch S2 (just move the cable on the Switch side) and when configuring Switch S2 to change the hostname to S2.

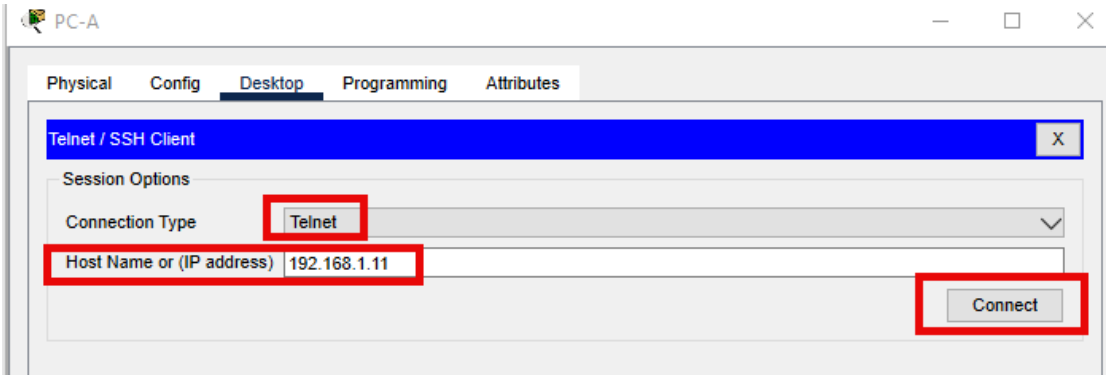
Part 4: Verify Basic Switch Settings – remote access

You will now use Telnet to remotely access the switch S1 using the SVI management address. Telnet is not a secure protocol. However, you will use it in this lab to test remote access. All information sent by Telnet, including passwords and commands, is sent across the session in plaintext. In subsequent labs, you will use SSH to remotely access network devices.

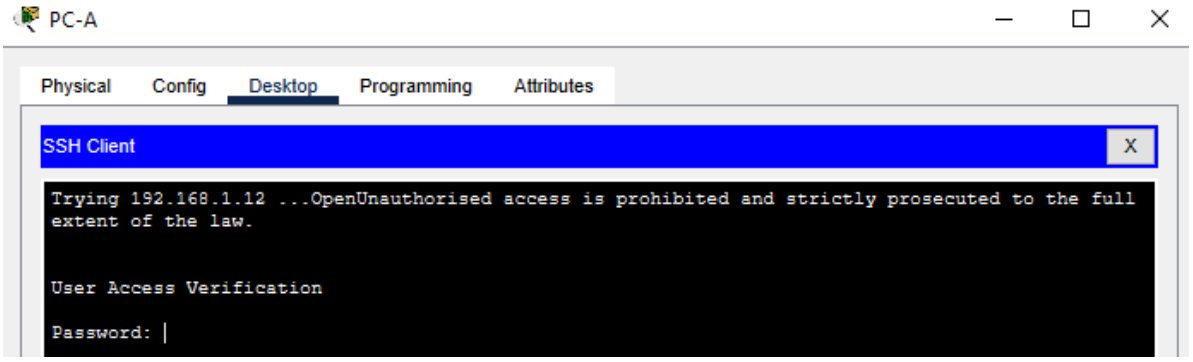
1. Open a new Telnet/SSH window in on PC-A. You can find this on its desktop (scroll down).



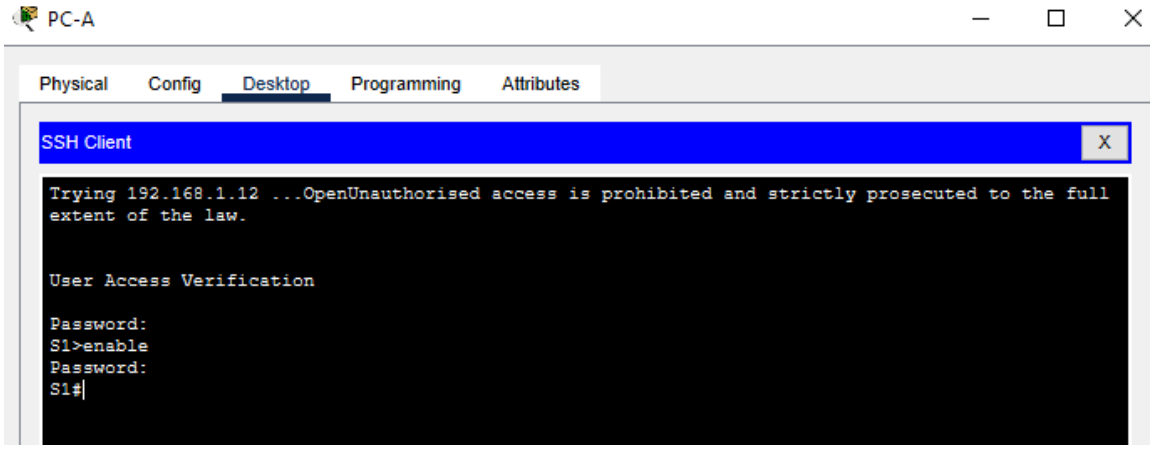
2. Select Telnet and enter the IP address in the Hostname field to connect to S1 via the SVI management address. Click Connect. The password is **cisco**.



3. In the diagram below, note the banner message at the top before the password prompt:



4. After entering the **cisco** password, you will be at the user EXEC mode prompt. Type **enable** at the prompt.Enter the **class** password to enter privileged EXEC mode and issue a **show run** command. Note: The text that you enter for your password will not display any characters to improve security



5. You have now successfully connected to the Switch via Telnet. If you did not reach this stage perform troubleshooting until this step is successful.
6. Confirm the commands that you have entered by showing the running configuration.
7. Repeat Steps 1 to 4 for Switch S2.

Reflection

What could prevent a ping from being sent between the PCs?

_____

Note: It may be necessary to disable the PC firewall to ping between PCs.

Part 5: Submit your Packet Tracer file

When you have completed the configuration activities in your Packet Tracer file, please save it as a .pka file (do not save as .pkz), and submit it using the Lab 2 submission point. By submitting it you will be able to become familiar with the submission process used for later Packet Tracer Assessments. We may also use your submissions to give feedback on if your name / student number has been correctly assigned to the file.

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